

BOOST: Bottleneck-Optimized Scalable Training Framework for Low-Rank Large Language Models

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Problem: Low-rank bottleneck LLMs save FLOPs and memory while preserving model quality, but vanilla tensor parallelism loses those performance gains at scale.

Key idea: BOOST restores scaling efficiency by **communicating** at narrow bottlenecks and **sharding** along hardware-friendly dimensions.

Improvement: End-to-end speedup: 1.46-1.91x over full-rank TP and 1.87-2.27x over vanilla low-rank TP.

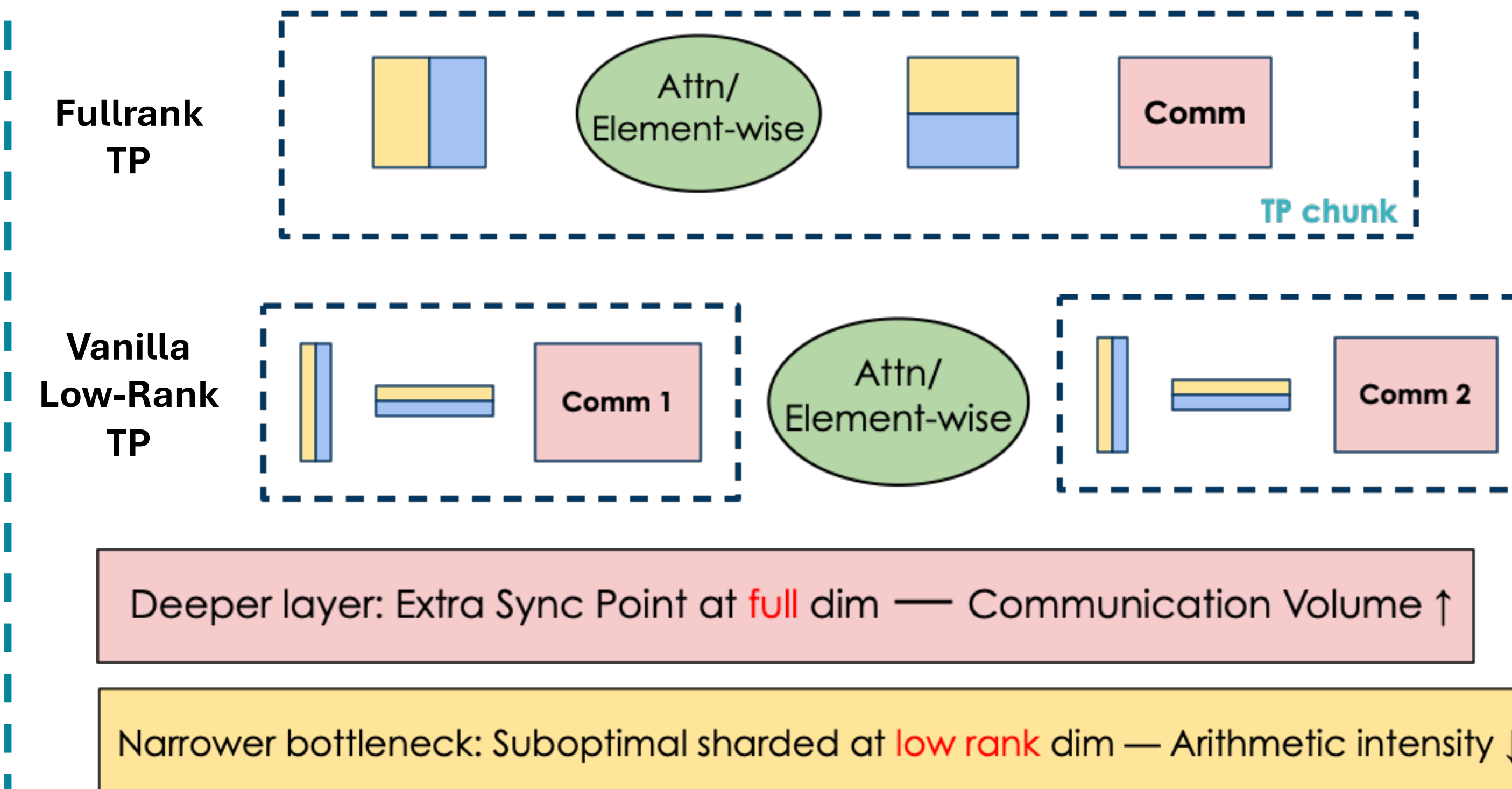
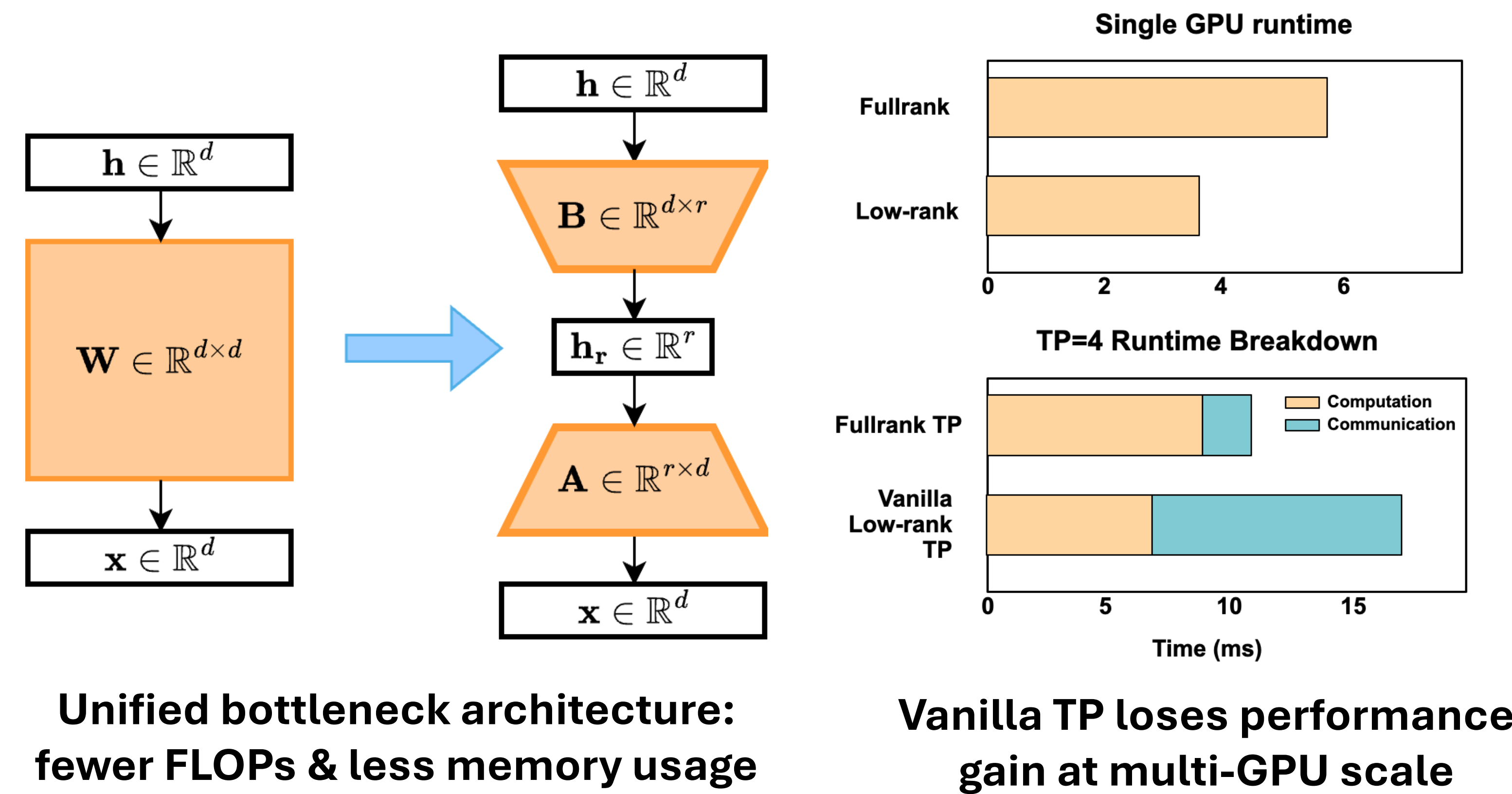


Code Repo

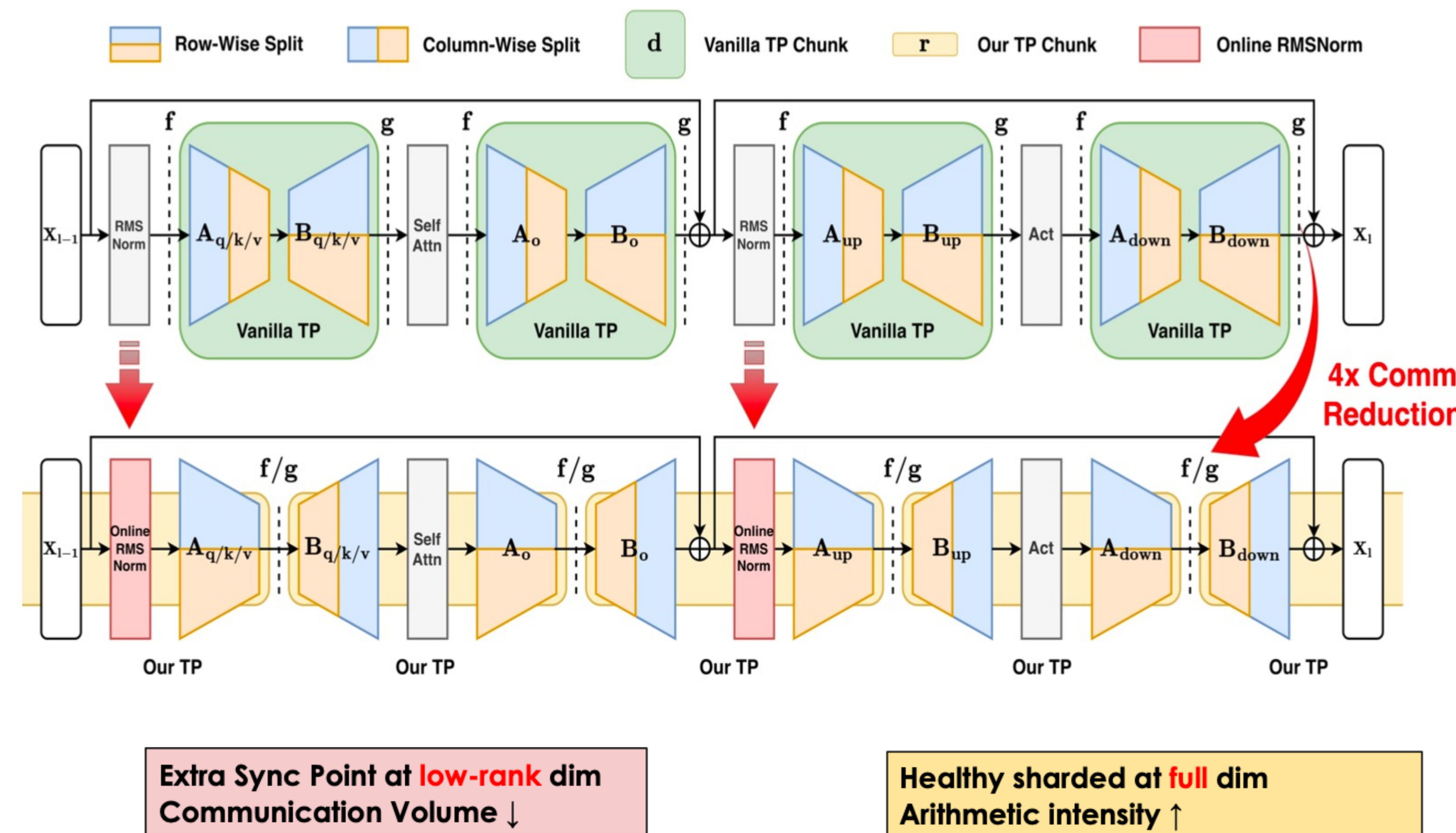


Paper

Background



Bottleneck-aware Tensor Parallelism (BTP) Design



System Optimization

#1: Online RMSNorm

$$\text{RMSNorm}(\mathbf{X}) = \frac{\mathbf{X} \cdot \gamma}{\text{RMS}(\mathbf{X})},$$

$$\text{with } \text{RMS}(\mathbf{X}) = \sqrt{\frac{1}{d} \sum_{j=1}^d \mathbf{X}_{:,j}^2} + \epsilon.$$

Latency bound comm kernel call

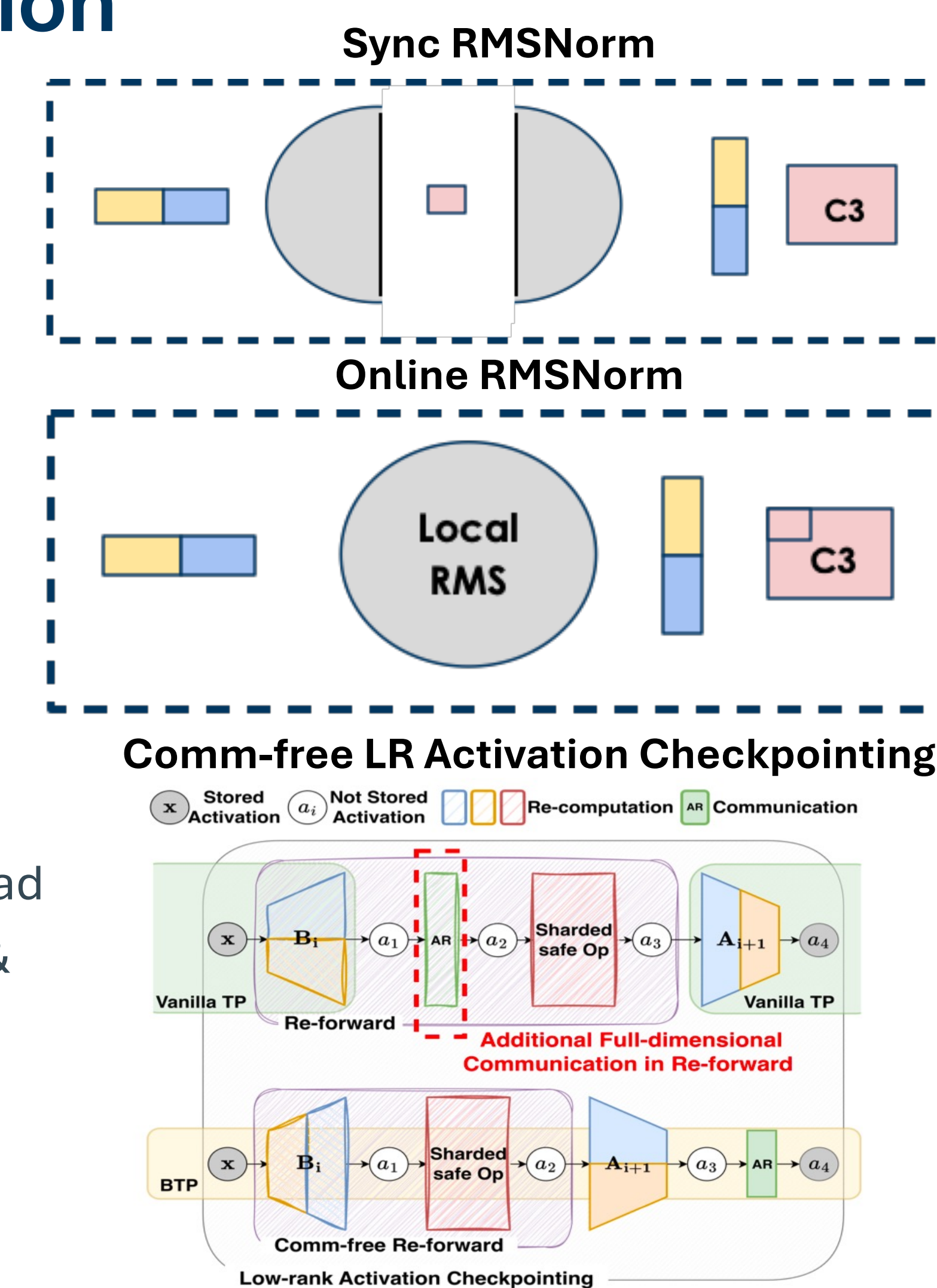
$$\sum_i \frac{W_i X_i}{\text{RMS}(\mathbf{X})} = \left(\sum_i \frac{W_i X_i}{\text{RMS}(X_i)} \right) \cdot \frac{1}{\text{RMS}(\mathbf{X})}$$

Tradeoff: Extra computation & Less communication call

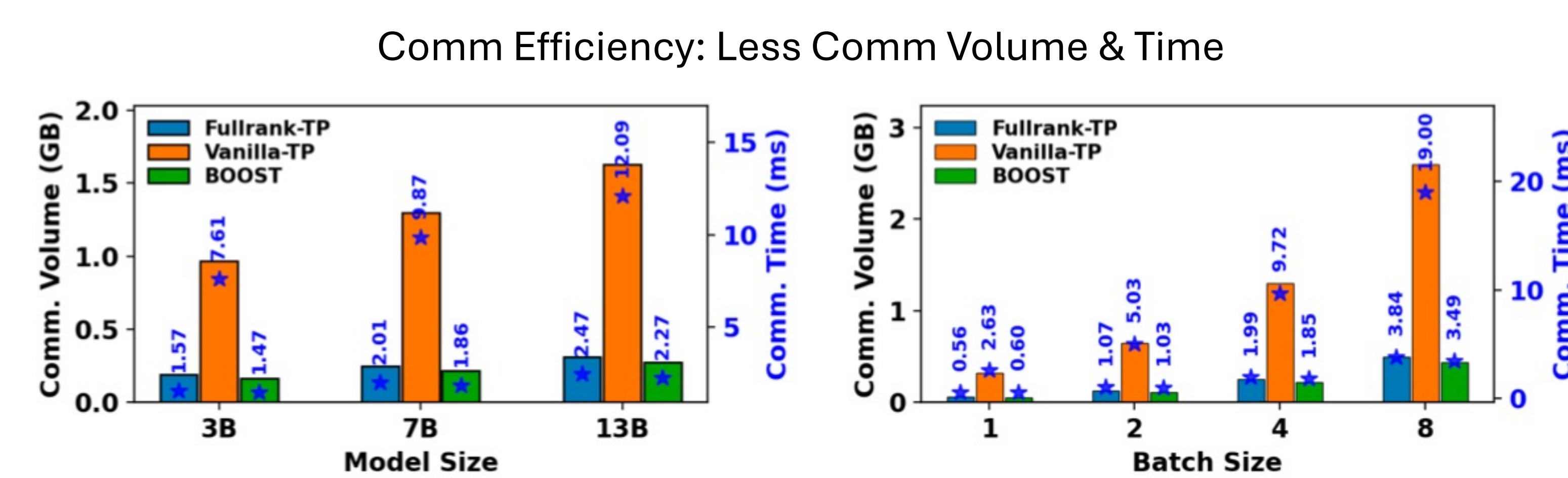
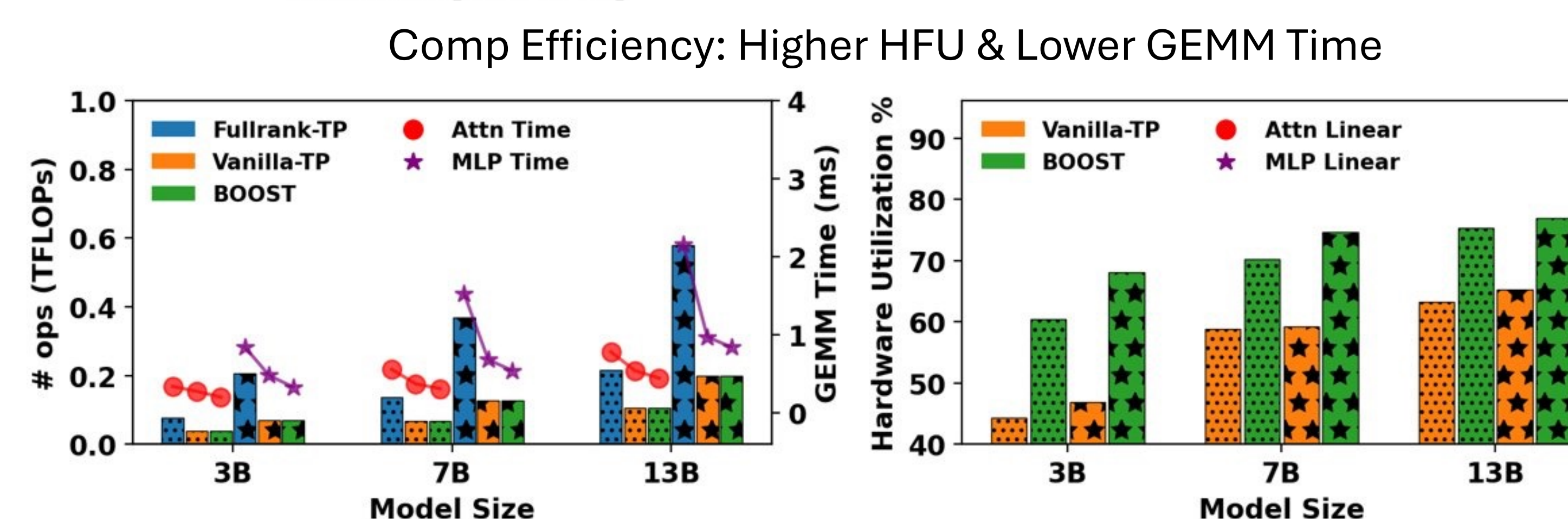
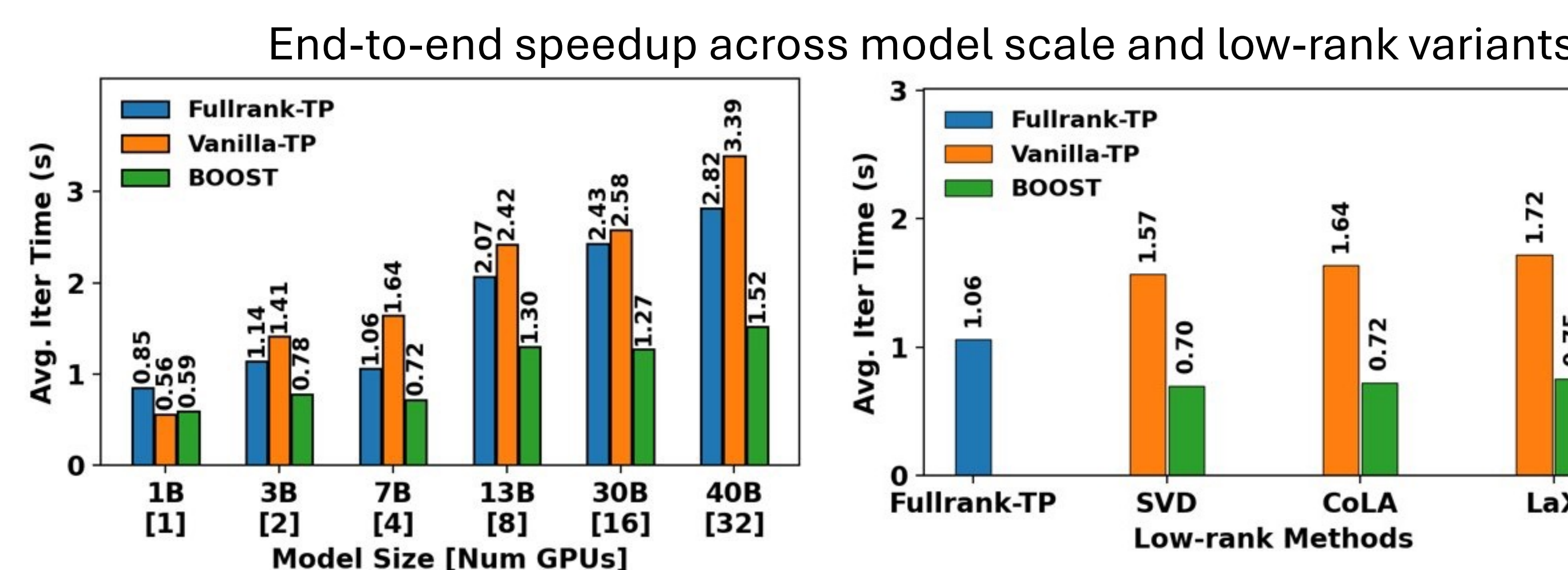
#2: Linear Layer Grouping

- Lower kernel launch overhead
- Higher arithmetic intensity & Comm bandwidth

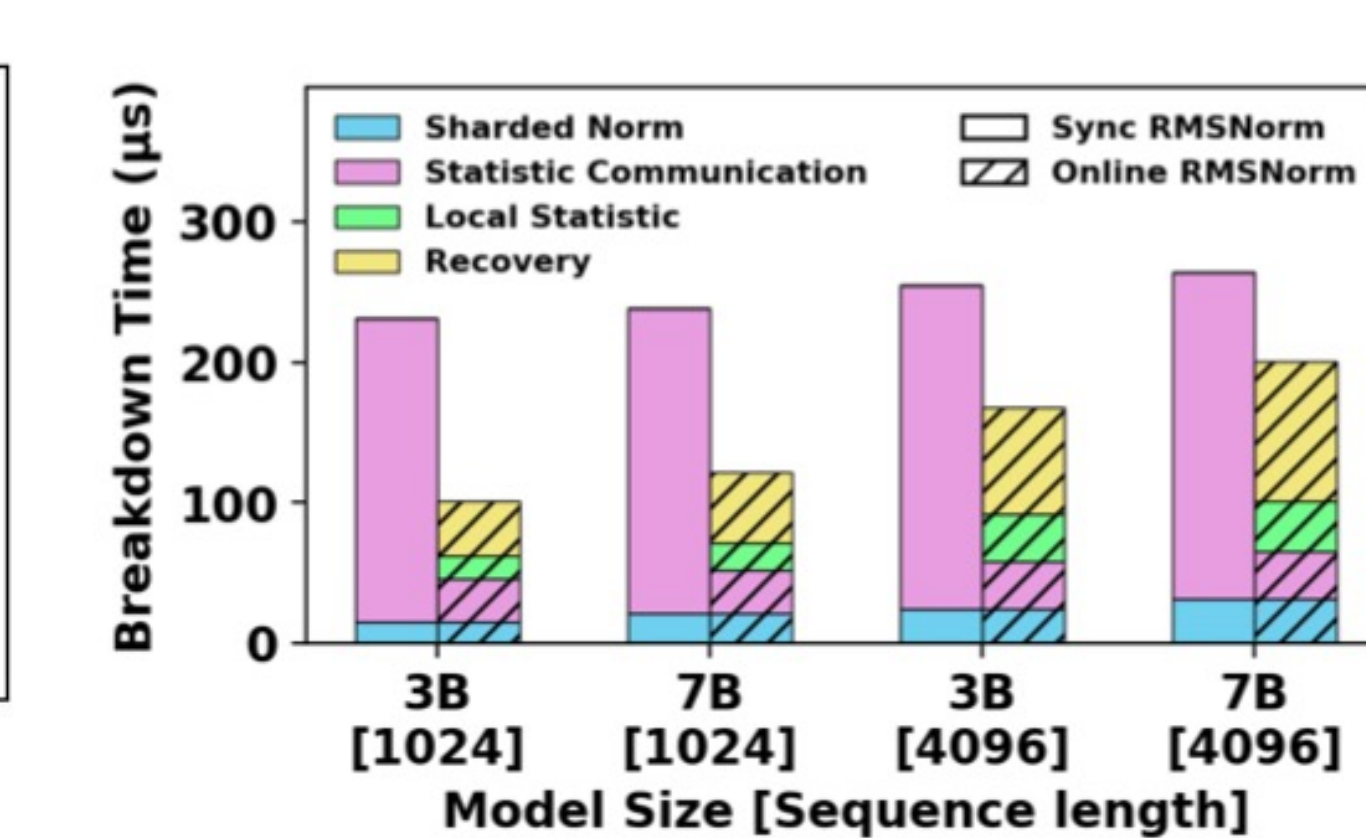
#3: Comm-free Low-rank Activation Checkpointing



Evaluation



Online RMSNorm Efficiency



Layer Grouping Speedup

Block / Kernel	bz=1			bz=4		
	Non.	Grouped	Speed	Non.	Grouped	Speed
MLP1 Comp	355	292	1.22×	1115	1082	1.03×
MLP1 Comm	266	218	1.22×	620	580	1.07×
QKV Comp	391	255	1.53×	939	877	1.07×
QKV Comm	406	288	1.41×	981	806	1.22×
Layerwise Total	2773	2395	1.16×	7577	7266	1.04×